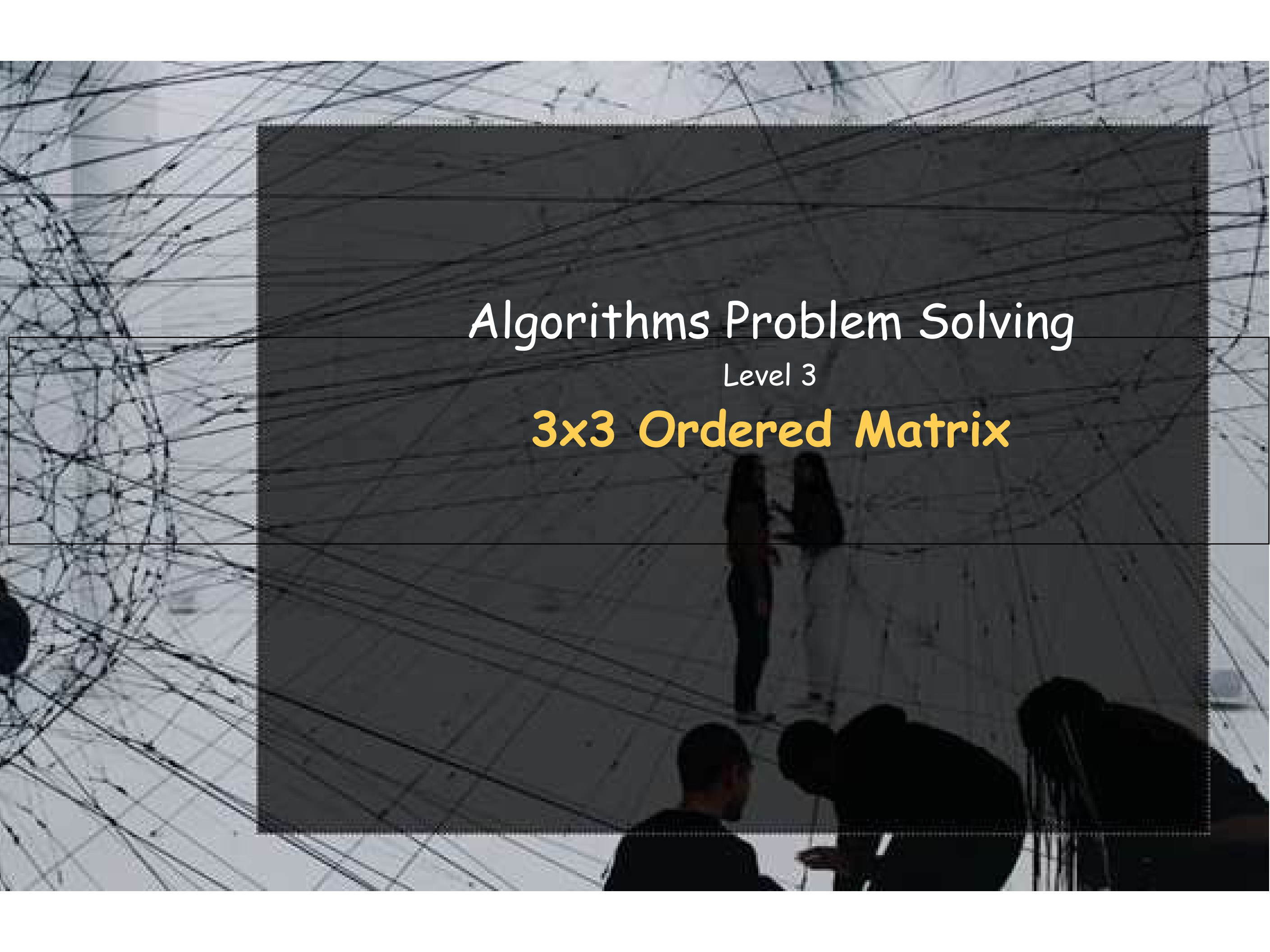


# Algorithms Problem Solving

## Level 3

### 3x3 Ordered Matrix



# Algorithms Problem Solving

Level 3

## 3x3 Ordered Matrix

# Definition

 3x3 Ordered Matrix

 Program fills a 3x3 matrix with ordered numbers from 1 to 9, then prints the matrix in a formatted way.

# Explanation

## Ordered Numbers

 The program uses a counter that starts at 1 and increases by 1 each time an element is filled.

 The elements are filled row by row, from left to right and from top to bottom.

## Two-Dimensional Matrix

 The matrix is two-dimensional and consists of rows and columns.

## setw()

 setw() from <iomanip> is used to format the numbers when displaying the matrix.

## Procedures

1. Define FillMatrixWithOrderedNumber() to fill the matrix with sequential numbers.
2. Define PrintMatrix() to print the matrix in a formatted table.
3. In main(), create a 3×3 matrix.
4. Call FillMatrixWithOrderedNumber(matrix, 3, 3).
5. Call PrintMatrix(matrix, 3, 3).

## 💡 Example

The following is a 3x3 ordered matrix:

|   |   |   |
|---|---|---|
| 1 | 2 | 3 |
| 4 | 5 | 6 |
| 7 | 8 | 9 |

➡ Result

The matrix is filled with consecutive numbers from 1 to 9.

## Note

- ◆ The number of rows and columns can be changed by using user input with cin.
- ◆ The variable num is used to keep track of the sequential numbers being added to the matrix.
- ◆ No random number generation is needed because the values are created in a fixed sequential order.

## ✦✦ Summary

- ✓ Populates a 3×3 2D vector sequentially with incrementing integers from 1 through 9.
- ✓ Utilizes nested loops to write entries systematically row-by-row and left-to-right.
- ✓ Formats tabular screen output consistently using standard `std::setw` formatting manipulators.

Thank you 😊